

The purpose of the proof-of-stake inclusion in its systems is meant to protect the currency from dangers that Bitcoin faces i.e. that if 51% of mining share is under a single control mechanism the currency can be manipulated (double spending) which would collapse the value of the currency. Since as mining rewards decrease with Bitcoin over time, there is a real possibility that mining will become highly monopolised allowing the possibility for a 51% attack. By having a hybrid system, new coin production is related to the how many coins are held by individuals i.e. someone with 1 percent of the Peercoins will create 1 percent of all proof-of-stake blocks. With an unlimited coin supply this will ensure that the cost of attempting a monopoly is a sufficient disincentive. Initially the system is built using proof-of-work implementation which gradually transitions to proof-of-stake as the former's difficulty increases.

Mastercoin

Mastercoin is one of the strangest cryptocurrency systems in use today. Mastercoin is positioned as the fifth largest currency with a market capitalisation of USD\$ 59,901,198 and interestingly the highest dollar exchange value next to Bitcoin.

Based on the Bitcoin protocol and not the currency model, Mastercoin is meant to “be used as a protocol layer, on top of which new currency layers with new rules can be built without changing the foundation”, according to the developer J. R. Willett in his whitepaper - “The Second Bitcoin Whitepaper”. It uses the already active Bitcoin protocol in order to create its own currency.

Mastercoin does not run on its own network or create new blocks or invite miners to dedicate power to earn coins. Instead its focused on creating new networks of currencies, commodities and securities using the Bitcoin protocol that is already in place. It uses the Bitcoin blockchain to store all its transactions and allows users with a host of new features such as the ability to create their own currency on the Mastercoin network.

The network functions as a decentralised exchange that negotiates between any two currencies by carrying orders for currencies on the Bitcoin blockchain and pairing it on the peer-to-peer network with a suitable pair instantly, without involvement from the offering party. Simply put, an individual could decide to sell 10 BTC at the rate of 1000 LTC for every 1 BTC and the offer is matched by someone else on the network by Mastercoin who is willing to make the exchange at that price.